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# Create Table Construct

- An SQL relation is defined using the **create table** command:

  **create table** *r*
  ($A_1$ $D_1$, $A_2$ $D_2$, ..., $A_n$ $D_n$,
  (integrity-constraint$_1$),
  ...,
  (integrity-constraint$_k$))

  - *r* is the name of the relation
  - each $A_i$ is an attribute name in the schema of relation *r*
  - $D_i$ is the data type of values in the domain of attribute $A_i$

- Example:

  **create table** *instructor* (
  **create table** *instructor* (
  *ID* **char**(5),
  *name* **varchar**(20),
  *dept_name* **varchar**(20),
  *salary* **numeric**(8,2))

3.7

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Title Slide
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- Example:

```
create table instructor (
        ID          char(5),
        name        varchar(20),
        dept_name   varchar(20),
        salary      numeric(8,2))
```



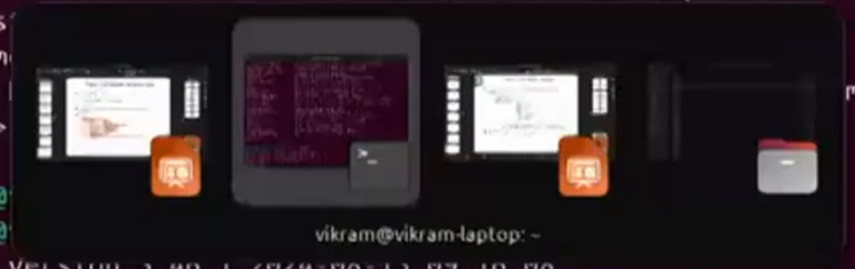
Slide 7 of 62 TextEdit: Paragraph 9, Row 16, Column 1 Title and Content 2.13 / 3.13 20.48 x 14.03

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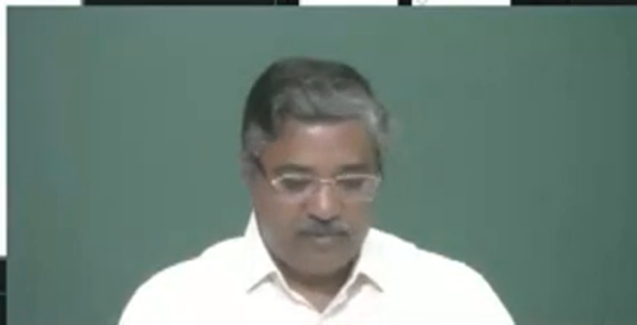
```
.schema ?PATTERN?        Show the CREATE statements matching PATTERN
.separator COL ?ROW?     Change the column and row separators
.session ?NAME? CMD ...  Create or control sessions
.sha3sum ...             Compute a SHA3 hash of database content
.shell CMD ARGS...       Run CMD ARGS... in a system shell
.show                    Show the current values for various settings
.stats ?ARG?             Show stats or turn stats on or off
.system CMD ARGS...      Run CMD ARGS... in a system shell
.tables ?TABLE?          List names of tables matching LIKE pattern TABLE
.timeout MS              Try opening locked tables for MS milliseconds
.timer on|off            Turn SQL timer on or off
.trace ?OPTIONS?         Output each SQL statement as it is run
.version                 Show source, library and compiler versions
.vfsinfo ?AUX?           Information about the top-level VFS
.vfslis
.vfsnam
.width                                                    mnar output
sqlite>

vikram@
vikram@
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite>
```




```
    dept_name  varchar(20),
    salary     numeric(8,2),
    primary key (ID),
    foreign key (dept_name) references department);
```

3.8



# String Operations

- SQL includes a string-matching operator for comparisons on character strings. The operator **like** uses patterns that are described using two special characters:
  - percent ( % ). The % character matches any substring.
  - underscore ( _ ). The _ character matches any character.
- Find the names of all instructors whose name includes the substring “dar”.

```
select name
from instructor
where name like '%dar%'
```

- Match the string “100%”

```
like '100 \%'  escape  '\'
```

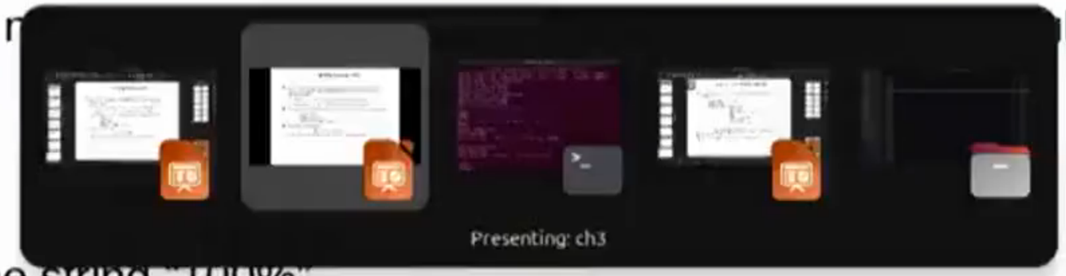
in that above we use backslash (\) as the escape character.





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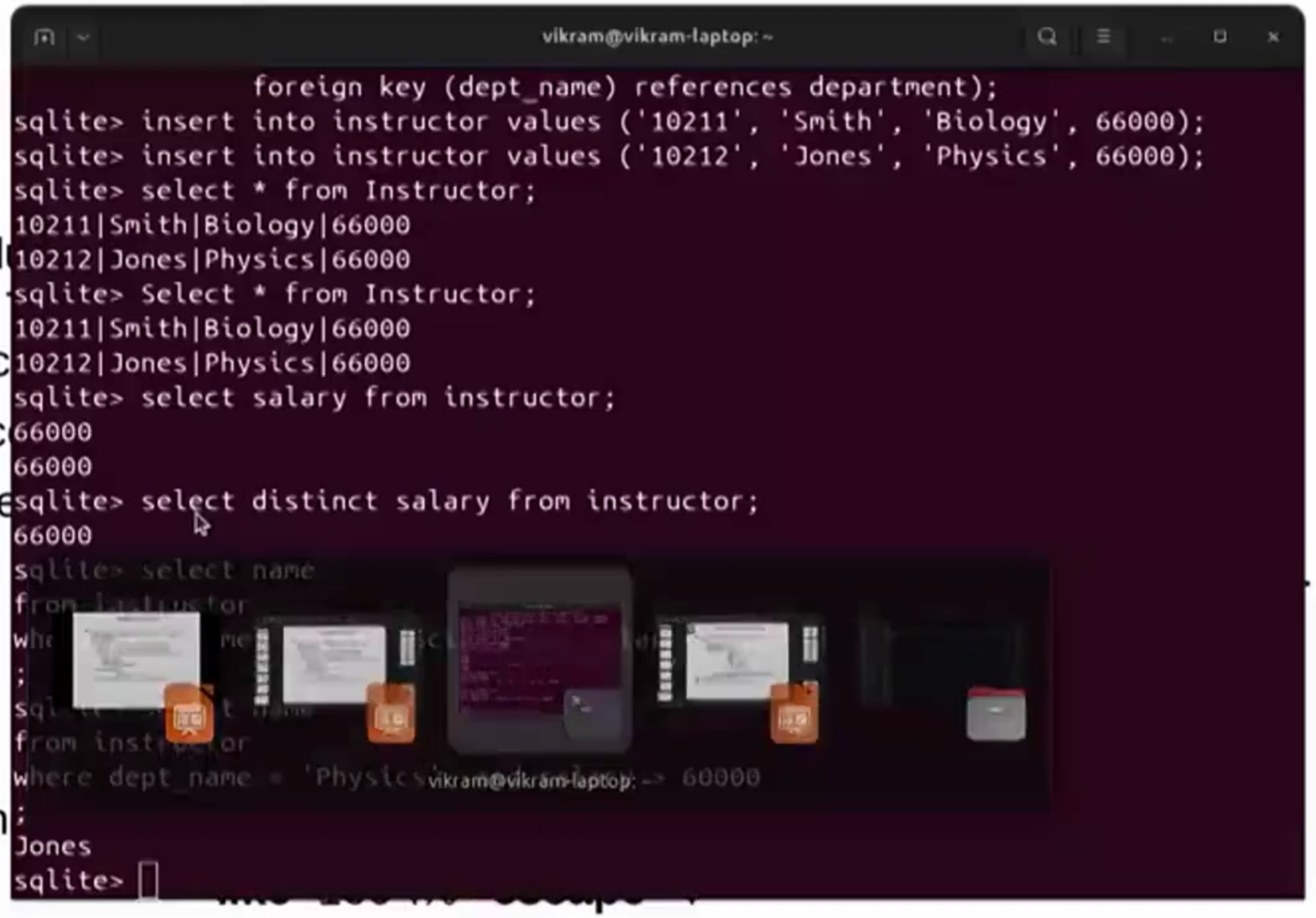



- Match the string “100%”

**like** '100 \%' **escape** '\'

in that above we use backslash (\) as the escape character.

- SQL incl
  strings.
  special c
  - perc
  - unde
- Find the
- Match th




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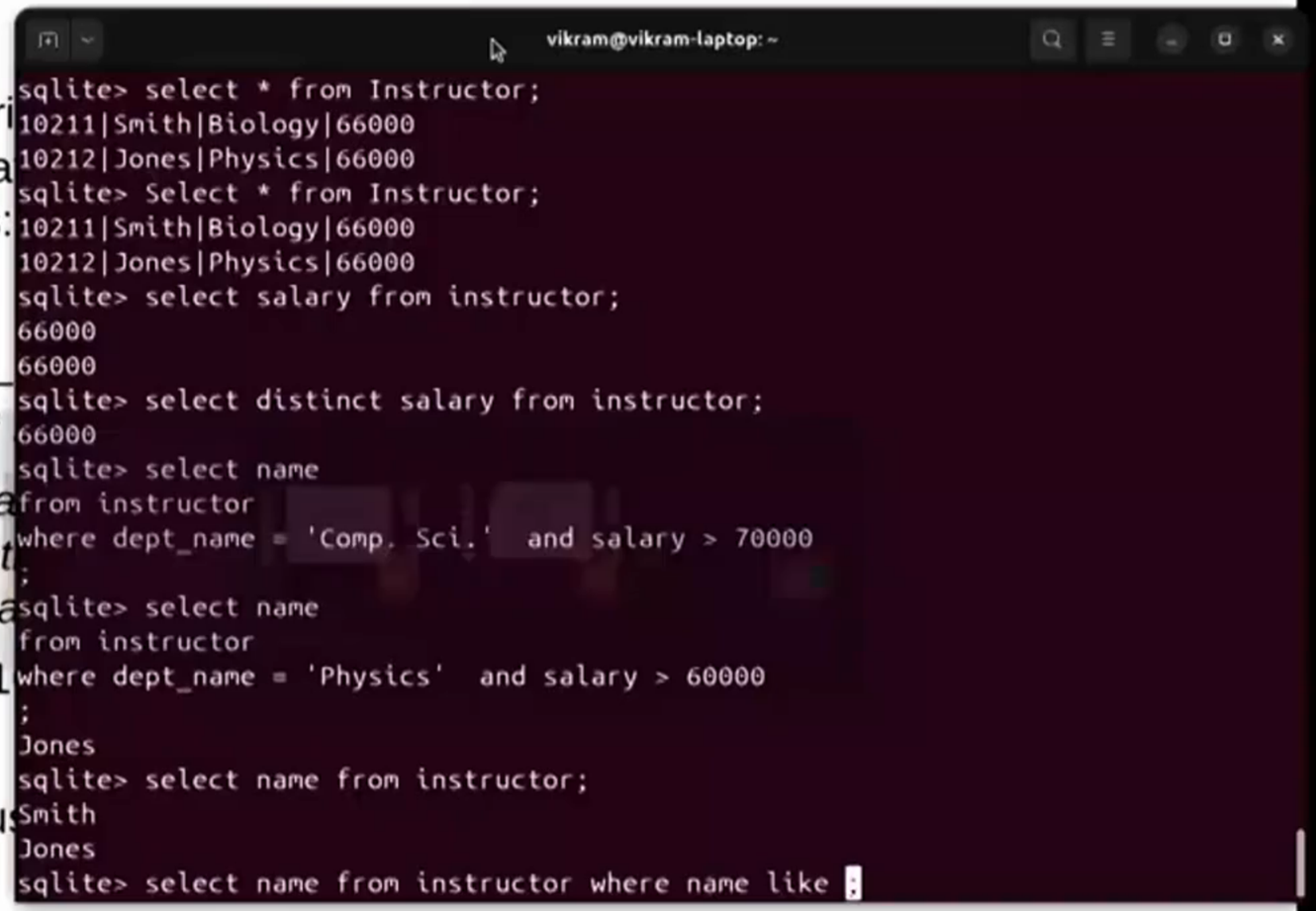
# String Operations

- SQL includes a stri
  strings. The opera
  special characters:
  - percent ( % ).
  - underscore ( _
- Find the names of

  **select** *na*
  **from** *inst*
  **where** *na*

- Match the string “1

in that above we u






# Aggregate Functions Examples

- Find the average s
  - **select avg** (*sa*
    **from** *instructo*
    **where** *dept_na*
- Find the total number of instructors who teach a course in the Spring 2018 semester
  - **select count** (**distinct** *ID*)
    **from** *teaches*
    **where** *semester* = 'Spring' **and** *year* = 2018;
- Find the number of tuples in the *course* relation
  - **select count** (*)
    **from** *course*;

```
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Smith
Jones
sqlite> select name from instructor where name="Jones";
Jones
sqlite> select name from instructor where name="J";
sqlite> select name from instructor where name like "J%";
Jones
```

